# Op-amps, differential amps, power amps

Cheat sheet · Day 21 · Fri 13 Nov · pack E13

## Reference-sheet lines

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DIFF / CMRR
 Vd = Vi1 - Vi2     Vc = (Vi1 + Vi2)/2     Vo = Ad*Vd + Ac*Vc
 CMRR = Ad/Ac       CMRR(dB) = 20 log10(Ad/Ac)    ratio = 10^(dB/20)
 Vo = Ad*Vd*(1 + Vc/(CMRR*Vd))          (use the ratio, not dB)
 Measure: Ad = Vo/Vd (Vc=0);  Ac = Vo/Vc (inputs tied)
BJT DIFF PAIR
 IE(tail) = (VEE - 0.7)/RE ; IC = IE/2 ; VC = VCC - IC*RC ; re = 26mV/IC
 single-ended Av = RC/(2re) ; double-ended Ad = RC/re ; Ac ~ RC/(2RE)
 Current mirror: IX = (Vsupply - VBE)/RX, I_out ~ IX
IDEAL OP-AMP: no input current; Vplus = Vminus (neg. feedback)
 Inverting  Vo = -(Rf/R1)V1         Non-inv  Vo = (1 + Rf/R1)V1
 Follower   Vo = V1                 Summer   Vo = -Rf*sum(Vk/Rk)
 Cascade: multiply stage gains, keep signs; check |Vo| < rails
POWER: eta = Po(ac)/Pi(dc); PQ = Pi - Po;  P = Irms^2 R = Ip^2 R/2 = Ipp^2 R/8
 Class A series: IB=(VCC-0.7)/RB, ICQ=bIB, VCEQ=VCC-ICQ*RC
   Pi = VCC*ICQ ; Po = Ic(rms)^2*RC ; Ic,pk(max) = min(ICQ, VCC/RC - ICQ)
   centred: Po,max = VCC^2/(8RC), eta max 25%
 Class A transformer: RL' = a^2 RL, a = sqrt(RL'/RL) ; eta max 50%
   Po = (VCEmax-VCEmin)(ICmax-ICmin)/8 ; Pi = VCC*ICQ
 Class B: Ip = VL(p)/RL ; Pi = 2VCC*Ip/pi ; Po = VL(p)^2/(2RL)
   eta = (pi/4)(VL(p)/VCC), max 78.5% ; P2Q = Pi - Po (both), PQ = P2Q/2
   max: Po = VCC^2/(2RL), Pi = 2VCC^2/(pi RL)
   P2Q,max = 2VCC^2/(pi^2 RL) at VL(p) = 0.636 VCC
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